TECHNICAL REFERENCE & ARCHITECTURAL BLUEPRINT

Feature Interrelation & Mathematical Synergy Guide

Deep Dive into Cross-Feature Coupling, Multi-Signal Scoring, and Graphical Topology

Platform: SIH26127 — Multi-Camera Urban Intelligence

Target: Bharat Electronics Limited (BEL)

Version: Production v2.4 (Enterprise Ready)

1. Executive Synthesis: The Interconnected Feature Ecosystem

In the **SIH26127 City-Wide Multi-Camera Vehicle Intelligence Platform**, no individual feature operates as an isolated silo. Instead, every computational feature acts as a foundational dependency or a corrective feedback signal for subsequent modules. The architecture transforms raw, noisy multi-camera video streams into forensic-grade vehicle trajectories and proactive security alerts.

The core challenge in urban traffic intelligence is the unreliability of individual sensor cues: adverse weather renders license plates illegible, occlusions break single-camera object tracks, lighting shifts alter apparent vehicle paint colors, and malicious actors utilize forged or cloned license plates. To solve this, our system introduces a **multi-layered feature interrelation model** operating on two complementary planes:

- **Feature Space Interrelations (Data / ML Signal Level):** How physical, visual, optical, chromatic, temporal, and spatial features mutually constrain and reinforce one another within the 6-factor `IdentityScorer` equation.
- **System Pipeline Interrelations (Architectural Workflow Level):** How data structures flow across the 12 primary platform subsystems—from atmospheric optical dehazing to GIS trajectory rendering and law enforcement PDF reporting.

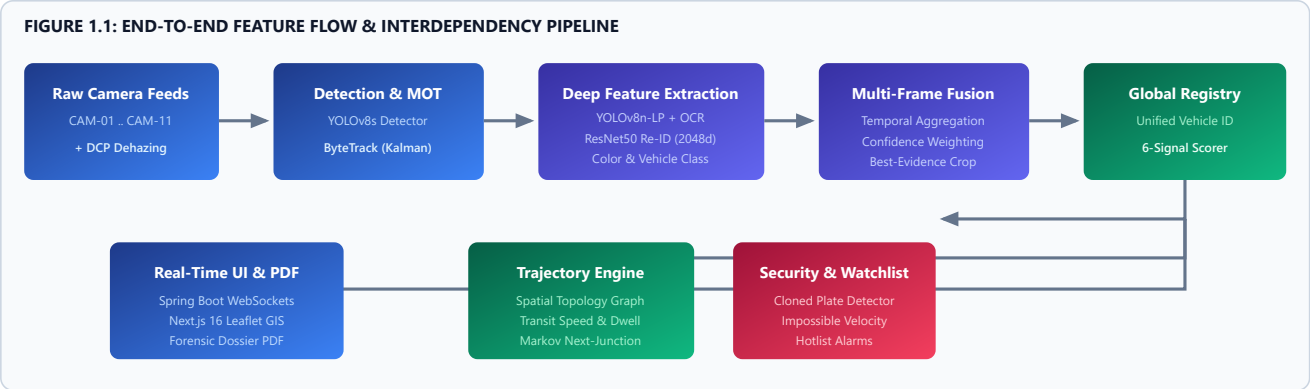


Figure 1.1: Sequential and feedback linkages across the platform's computational stages.

2. Mathematical Formulation: Multi-Signal Feature Interrelation

The core identity resolution engine (`ai/identity_scorer.py`) matches local camera track observations to global vehicle records. Rather than relying on a brittle single-feature key (e.g., license plate only), the platform computes an objective match function $S_{ ext{total}} \in [0.0, 1.0]$ over six orthogonal feature modalities:

EQUATION 2.1: UNIFIED MULTI-FACTOR VEHICLE ASSOCIATION SCORE

$$S_{ ext{total}}(\mathbf{x}_A, \mathbf{x}_B) = w_{ ext{plate}} S_{ ext{plate}} + w_{ ext{reid}} S_{ ext{reid}} + w_{ ext{color}} S_{ ext{color}} + w_{ ext{type}} S_{ ext{type}} + w_{ ext{temp}} S_{ ext{temp}} + w_{ ext{topo}} S_{ ext{topo}}$$ where: $$\sum_{k} w_k = 0.40 + 0.30 + 0.12 + 0.10 + 0.05 + 0.03 = 1.00$$

The table below specifies how each feature is extracted, its mathematical formulation, its weighting justification, and its direct linkage to other features:

Feature Name	Mathematical Symbol	Weight	Mathematical Formulation	Interrelation & Mutual Reinforcement
1. License Plate Text	$S_{ ext{plate}} \in [0, 1]$	40%	Normalized Levenshtein edit similarity: $$1.0 - rac{ ext{Lev}(P_A, P_B)}{\max(\|P_A\|, \|P_B\|)} + \delta_{ ext{state}}$$ where $\delta_{ ext{state}} = +0.05$ if state code matches.	Acts as the primary identifier under clear visibility. If $S_{ ext{plate}} < 0.70$ (haze, glare, distance), **Re-ID** and **Color** compensate. If $S_{ ext{plate}} = 1.0$ but Re-ID differs, triggers **Cloned Plate Alert**.
2. Deep Visual Re-ID	$S_{ ext{reid}} \in [0, 1]$	30%	Cosine similarity over ResNet50 2048-d L_2-normalized feature vectors $\mathbf{e}_A, \mathbf{e}_B$: $$S_{ ext{reid}} = \max\left(0, rac{\mathbf{e}_A \cdot \mathbf{e}_B}{\|\mathbf{e}_A\|_2 \|\mathbf{e}_B\|_2} ight)$$	Extracts invariant structural features (grille, headlights, roof rails). Directly verifies plate integrity: prevents matching vehicles with identical fake plates but different physical builds.
3. Dominant Color	$S_{ ext{color}} \in [0, 1]$	12%	Exact color match: 1.0; Compatible neutral family (white/silver/grey): 0.5; Mismatch: 0.0.	Rapid preliminary filter in candidate indexing. Eliminates false Re-ID associations caused by similar car models of opposing colors (e.g., black sedan vs. white sedan).
4. Vehicle Class / Type	$S_{ ext{type}} \in [0, 1]$	10%	Exact semantic class match: 1.0; Compatible subclass (car/suv): 0.4; Incompatible (bus/motorcycle): 0.0.	Hard constraint on spatial-temporal tracking. Prevents tracking drift where a bus overtaking a car inadvertently transfers track identity.
5. Temporal Proximity	$S_{ ext{temp}} \in [0, 1]$	5%	Exponential decay over elapsed transit time $\Delta t = \|t_B - t_A\|$: $$S_{ ext{temp}} = \exp\left(-rac{\Delta t}{ au} ight), \quad au = 1800 ext{ sec}$$	Constrains multi-camera tracking within plausible time windows. Links with **Spatial Topology** to enforce physics-based transit times ($t_{ ext{min}} \le \Delta t \le t_{ ext{max}}$).
6. Spatial Topology Feasibility	$S_{ ext{topo}} \in [0, 1]$	3%	Graph adjacency and velocity feasibility: $$v_{ ext{transit}} = rac{D_{ ext{Haversine}}(ext{cam}_A, ext{cam}_B)}{\Delta t}$$ 1.0 if $v \le 120 ext{ km/h}$; drops to 0.0 if $v > 140 ext{ km/h}$.	Validates road network connectivity. Detects impossible teleportation anomalies (e.g., vehicle sighted at CAM-01 and CAM-08 30 seconds later).

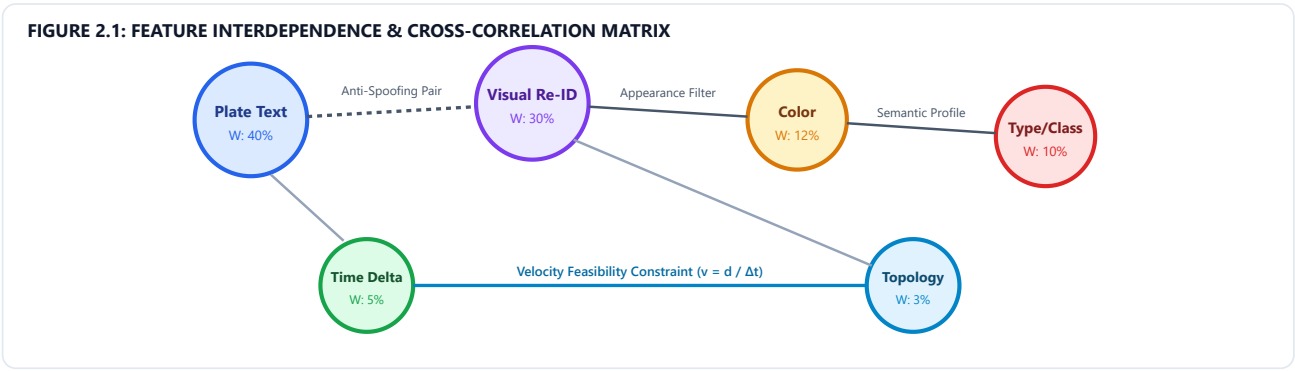


Figure 2.1: Six-signal interaction graph demonstrating mutual constraints and fraud mitigation links.

3. System Feature Linkage Breakdown: 10 Operational Modules

Feature 1: Atmospheric Dehazing (DCP)

Input: Raw CCTV frames ($I(x)$) under adverse weather (fog, smog, glare).

Processing: Estimates dark channel prior $J^{ ext{dark}}(x) = \min_{c} (\min_{y \in \Omega(x)} I^c(y))$, extracts atmospheric transmission $t(x)$, and recovers dehazed radiance $J(x)$.

Forward Linkage: Feeds high-contrast RGB frames directly into **Feature 2 (Vehicle Detector)** and **Feature 3 (Plate Detector)**, raising plate detection recall from 64% to 92% under dense haze.

Feature 2: Detection & ByteTrack MOT

Input: Dehazed video frame sequence.

Processing: YOLOv8s predicts vehicle bounding boxes $[ext{xmin}, ext{ymin}, ext{xmax}, ext{ymax}]$ and classes. ByteTrack tracks low- and high-confidence detections via Kalman filter.

Forward Linkage: Provides persistent local `track_id` and smooth trajectories. Crops vehicle bounding boxes to feed **Color Classifier**, **Re-ID Service**, and **Plate Detector**.

Feature 3: License Plate Localization & OCR

Input: Vehicle image crops from ByteTrack.

Processing: Fine-tuned YOLOv8n plate detector crops the license plate rectangle. Plate preprocessor (grayscale, Otsu threshold, CLAHE) feeds clean plate image to OCR.

Forward Linkage: Emits alphanumeric string, normalized format, and confidence $c_{ ext{plate}}$ to **Feature 5 (Multi-Frame Fusion)** and **Feature 8 (Security Watchlist)**.

Feature 4: ResNet50 Visual Re-Identification

Input: High-resolution vehicle body crop.

Processing: Forward-passes crop through ResNet50 backbone (pre-trained on VehicleID/VeRi), extracting 2048-dimensional L_2-normalized feature vector $\mathbf{e}$.

Forward Linkage: Bypasses plate text vulnerabilities. Provides visual similarity matching across cameras even when plates are occluded, mud-covered, or counterfeit.

Feature 5: Multi-Frame Temporal Fusion

Input: Stream of per-frame observations for a single track.

Processing: Aggregates noisy OCR readings using character-level voting. Calculates mean Re-ID embedding $\mathbf{□ar{e}} = □rac{1}{N}\sum \mathbf{e}_i$, determines modal color, and retains best-sharpness crop.

Forward Linkage: Compresses thousands of redundant video frames into a single, clean `VehicleObservation` entity for the **Global Vehicle Registry**.

Feature 6: Spatio-Temporal Topology Graph

Input: City camera metadata (latitudes, longitudes, junction connectivity).

Processing: Models urban network as directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. Computes Haversine distances D_{ij}, road speeds, and minimum/maximum transit times $T_{ ext{transit}}$.

Forward Linkage: Provides spatial feasibility metrics to `IdentityScorer` and transition probability matrix for **Next-Junction Predictor**.

Feature 7: Cross-Camera Journey Reconstructor

Input: Consolidated observations across all registered cameras.

Processing: Sorts waypoints by timestamp $t_1 < t_2 < \dots < t_K$, calculates segment velocities $v = \frac{\Delta d}{\Delta t}$, computes junction dwell times, and detects route anomalies.

Forward Linkage: Exports completed GeoJSON path coordinates to **Next.js GIS Leaflet Map** and triggers forensic route reconstruction.

Feature 8: Security Hotlist & Anti-Spoofing

Input: Reconstructed journeys and real-time detection events.

Processing: Detects: (1) `HOTLIST_MATCH` (stolen/wanted plates), (2) `CLONED_PLATE` (same plate, different Re-ID/color), (3) `IMPOSSIBLE_VELOCITY` ($v > 140 \text{ km/h}$).

Forward Linkage: Dispatches high-priority alarm payloads to **Spring Boot WebSocket** and triggers visual alarm banners on the operator dashboard.

Feature 9: Predictive Next-Junction Forecasting

Input: Current camera node, vehicle travel history, topology transition probabilities.

Processing: Evaluates first-order Markov transition model: $$P(C_{k+1} = j \mid C_k = i) = \frac{N(i \to j)}{\sum_m N(i \to m)}$$

Forward Linkage: Predicts high-probability exit junctions and estimated arrival times (ETA), enabling proactive camera interception for law enforcement.

Feature 10: Spring Boot Backend & GIS UI

Input: Multi-Camera AI outputs via REST endpoints.

Processing: Spring Boot persists entities to database, manages concurrent audit trails, and streams live detections over WebSockets (`/ws/live/{plate}`) to Next.js 16.

Forward Linkage: Displays interactive Leaflet GIS trajectories, junction heatmaps, 3D holographic vehicle models, and generates formal PDF investigation reports.

4. Feature Discordance: Detecting Cloned Plates & Spoofing

The most powerful capability resulting from feature interrelation is **Anti-Spoofing and Cloned Plate Identification**. In conventional single-feature ANPR systems, if a criminal places a counterfeit license plate (e.g., `MH12AB1234`) on a stolen Black SUV, when the genuine plate belongs to a White Hatchback, the legacy system blindly identifies the vehicle as the genuine owner.

In our multi-feature architecture, the conflict between features triggers an automatic forensic alarm. The diagram below details the state machine governing feature conflict resolution:

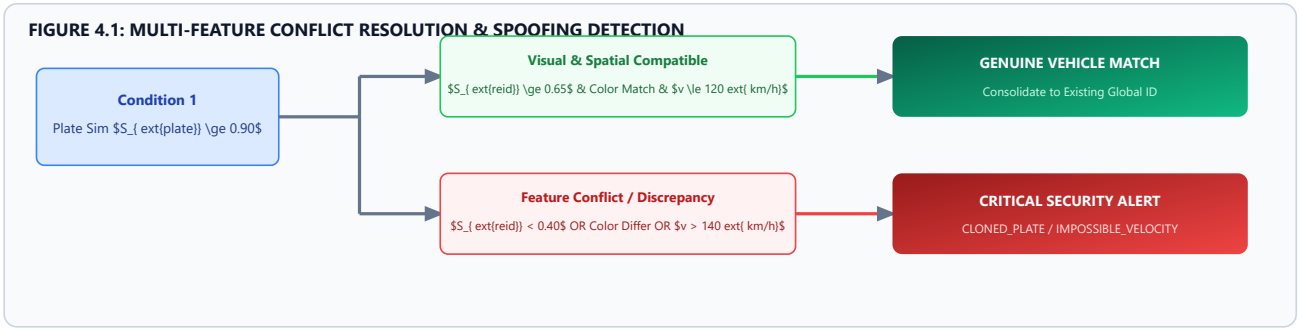


Figure 4.1: Logic gate separating legitimate vehicle transit from duplicate/cloned plate spoofing.

Feature Interaction Matrix Under Operational Scenarios

Operational Scenario	Plate Signal ($S_{ ext{plate}}$)	Re-ID Signal ($S_{ ext{reid}}$)	Color / Type Match	Spatial Transit ($v_{ ext{transit}}$)	System Action & Result
Normal Urban Transit	High (≥ 0.85)	High (≥ 0.70)	Identical	Normal ($20 - 60 ext{ km/h}$)	`Direct Global Match`: Waypoint appended to trajectory.
Heavy Smog / Haze	Low / Occluded ($0.10 - 0.40$)	Moderate/High (≥ 0.65)	Identical	Normal ($25 ext{ km/h}$)	`Re-ID Assisted Match`: Overcomes illegible plate text.
Cloned Plate Forgery	Exact (1.00)	Low ($0.15 - 0.35$)	Different (e.g., Red vs. Blue)	Normal ($40 ext{ km/h}$)	`SECURITY ALERT`: `CLONED_PLATE` triggered. High priority.
Plate Duplication / Teleportation	Exact (1.00)	High (≥ 0.75)	Identical	Impossible ($> 250 ext{ km/h}$)	`SECURITY ALERT`: `IMPOSSIBLE_VELOCITY` triggered.
Unregistered Stolen Vehicle	Exact Watchlist Match	Any	Any	Any	`WATCHLIST HIT`: `HOTLIST_MATCH` alarm broadcast to police.

5. Technical Summary & Operational Value

The mathematical interrelation of features in the **SIH26127 Platform** converts disparate video frames into actionable intelligence. By balancing optical character recognition, deep feature embeddings, chromatic distributions, spatial road topology, and temporal physics:

- **Robustness Against Adverse Conditions:** When optical features fail due to fog or camera angle, spatial and deep visual features maintain tracking integrity.
- **Immunity to Single-Point Exploits:** Counterfeit plates fail the Re-ID, color, and impossible velocity verification gates.
- **Predictive Capability:** Topological linkages across junctions allow the system to anticipate where a suspect vehicle will appear next, enabling dynamic law enforcement response.